

ROI Lens

Marketing Spend Optimization

A Data-Driven Framework for Marketing Attribution,
Campaign Optimization & Budget Reallocation

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Business Problem & Dataset Overview

Current State

- ₹100 Cr annual marketing spend across 5 channels and 50 campaigns
- 100,000 customers generating 566,510 marketing touchpoints
- 5,498 recorded purchases across the customer journey

Challenge

- Last-click attribution overcredits conversion channels and undervalues awareness channels
- Budget allocation is not fully aligned with channel performance
- Excessive customer exposures may be increasing marketing waste and acquisition costs

Objective

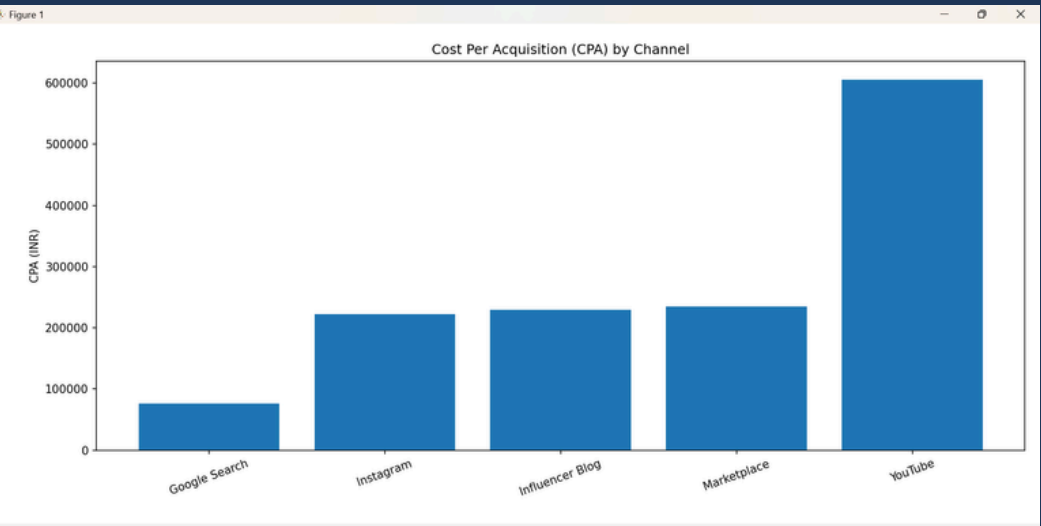
- Identify the true contribution of each marketing channel
- Optimize budget allocation using attribution and CPA insights
- Improve conversion efficiency and maximize marketing ROI

How can Nexus Consumer Brands reallocate a ₹100 Crore marketing budget to maximize conversions while reducing inefficient spend and preserving valuable awareness channels?

Channel Performance Analysis

Despite receiving a smaller share of the total budget, Google Search generated the highest conversion efficiency and the lowest CPA, making it the most effective acquisition channel.

Channel	Conversion Rate	CPA
Google Search	11.20%	₹76,280
Marketplace	4.40%	₹234,936
Instagram	3.70%	₹222,377
Influencer Blog	2.10%	₹229,337
YouTube	1.90%	₹605,959



Strategic Findings

- **Google Search is the strongest conversion driver**
 - Highest conversion rate (11.2%)
 - Lowest CPA (₹76,280)
 - Generated the highest number of purchases (2,661)
- **YouTube is the least efficient acquisition channel**
 - Highest CPA (₹605,959)
 - Lowest conversion rate (1.9%)
 - Requires nearly 8× higher spend per acquisition than Google Search
- **Budget allocation is not aligned with performance**
 - Google Search receives less funding than YouTube and Marketplace despite superior results
 - Indicates significant opportunity for budget reallocation

Google Search delivers the strongest marketing efficiency, generating the highest conversion rate at the lowest acquisition cost. Current budget allocation does not reflect channel performance, creating a significant opportunity for optimization.

User Behavior & Frequency Analysis

Customer Engagement Patterns:

- Average customer journey length: 5.67 touchpoints
- Median interactions: 4 touchpoints
- 75% of users interacted 5 times or fewer
- 1,952 users recorded more than 20 interactions without converting

Key Findings;

- ✓ Most customers convert within 4–6 interactions
- ✓ 75% of users convert within five touchpoints.
- ✓ 1.95% of users generated 20+ interactions without converting.
- ✓ Excessive exposure can contribute to ad fatigue and wasted spend

Recommended Frequency Cap 5–7 touchpoints per user

Metric	Current State	After Frequency Cap (5–7)
Total Touchpoints	566,510	383,726
Excess Interactions	182,784	Eliminated
Exposure Volume	100%	67.70%
Exposure Reduction	0%	32.30%
Frequency Policy	No Cap	5–7 Touchpoints

Strategic Impact

- Eliminates 182,784 low-value interactions
- Reduces total exposure volume by 32.3%
- Improves budget efficiency without increasing spend

Implementing a 5–7 touchpoint frequency cap can eliminate 182,784 low-value interactions and reduce marketing exposure by 32.3% without compromising conversion opportunities.

Attribution Analysis

Channels Under-Valued by Last-Click Attribution

Channel	First Click	Linear	Last Click
Google Search	1462	1672	2661
Influencer Blog	1147	1018	501
Instagram	988	992	873
Marketplace	987	993	1024
YouTube	914	823	439

Key Insights:

- Google Search remains the strongest conversion driver across all attribution models.
- Influencer Blog receives 129% more credit under First-Click Attribution (1,147 vs 501).
- YouTube receives 108% more credit under First-Click Attribution (914 vs 439).
- Last-Click Attribution systematically undervalues channels that initiate customer journeys.

Attribution analysis reveals that awareness channels contribute far more to customer acquisition than indicated by last-click metrics, highlighting the need for attribution-driven budget allocation.

₹100 Crore Budget Reallocation Strategy

Channel	Current (Normalized)	Proposed	Change
Google Search	₹19.9 Cr	₹35.0 Cr	+15.1 Cr
Instagram	₹19.0 Cr	₹22.0 Cr	+3.0 Cr
Marketplace	₹23.7 Cr	₹20.0 Cr	-3.7 Cr
Influencer Blog	₹11.3 Cr	₹13.0 Cr	+1.7 Cr
YouTube	₹26.1 Cr	₹10.0 Cr	-16.1 Cr
Total	₹100 Cr	₹100 Cr	0

Strategic Recommendations:

- Reallocate budget from low-value campaigns to high-performing conversion drivers
- Implement a 5–7 touchpoint frequency cap to eliminate inefficient exposures
- Adopt multi-touch attribution for future budget allocation decisions
- Continuously optimize campaign spend based on CPA and attribution performance

Proposed allocation based on CPA analysis, attribution insights, and frequency optimization findings.

The proposed ₹100 Cr allocation prioritizes channels that demonstrably drive customer acquisition while reducing inefficient spend and preserving critical awareness touchpoints.

Expected Business Impact

Improved Conversion Efficiency

- Increased investment in high-performing acquisition channels
- Greater focus on channels with proven conversion effectiveness
- Higher conversion volume without increasing total budget

Reduced Marketing Waste

- Frequency cap eliminates 182,784 low-value interactions
- Reduces customer overexposure by 32.3%
- Minimizes inefficient marketing spend

Better Budget Allocation

- Attribution-driven budgeting improves investment decisions
- Awareness and conversion channels evaluated more accurately
- Reduces reliance on misleading Last-Click metrics

Stronger Marketing ROI

- Budget redirected toward higher-value opportunities
- Improved acquisition efficiency across the marketing portfolio
- Maximizes ROI without increasing total marketing spend

By combining attribution-driven budgeting, frequency optimization, and strategic portfolio reallocation, Nexus Consumer Brands can improve marketing efficiency, reduce wasted spend, and maximize the return on its ₹100 Crore marketing investment.

Conclusion & Final Recommendations

Key Findings:

- ✓ Google Search emerged as the most efficient acquisition channel, delivering the highest conversion rate (11.2%) and the lowest CPA (₹76,280).
- ✓ Customer journeys exhibit diminishing returns beyond 5–7 touchpoints, with a frequency cap capable of eliminating 182,784 low-value interactions and reducing exposure by 32.3%.
- ✓ Multi-touch attribution revealed that awareness channels contribute significantly more to customer acquisition than indicated by Last-Click Attribution, highlighting the need for attribution-driven decision making.

Final Recommendations:

- 📌 Reallocate budget toward high-performing conversion channels.
- 📌 Implement a 5–7 touchpoint frequency cap to reduce marketing waste.
- 📌 Adopt multi-touch attribution for future budget allocation decisions.
- 📌 Continuously optimize campaign investments using CPA, attribution, and customer journey insights.

By aligning marketing investments with customer behavior and attribution insights, Nexus Consumer Brands can improve conversion efficiency, reduce wasted spend, and maximize returns from its ₹100 Crore marketing budget.